

Cancer Compass

A Continuously Updating Cancer Diagnostic Pathway Platform

What We Are Building

We are developing a hospital-facing research platform intended to help clinicians find the fastest safe path toward confirming or ruling out cancer.

Patients with unclear symptoms may undergo repeated appointments, tests, treatments, and referrals before receiving a definitive diagnosis. Each individual decision may be medically reasonable, but the overall diagnostic pathway can become fragmented, repetitive, or unnecessarily slow when information is spread across multiple visits and departments.

Cancer Compass would review the patient's longitudinal medical record and evaluate the medically appropriate tests, procedures, referrals, or monitoring options available at that moment. It would then identify the next step expected to reduce the most diagnostic uncertainty while considering safety, time, burden, cost, and previous work.

The platform would not diagnose cancer or automatically order tests. It would provide clinicians with a transparent, evidence-linked recommendation that they could accept, reject, or modify.

The long-term goal is to create a continuously updating diagnostic navigator that helps hospitals reduce unnecessary tests and referrals while helping the right patients reach definitive diagnosis earlier.

Research and Model Approach

We expect to study a longitudinal clinical decision-support system that analyzes information already collected throughout a patient's diagnostic journey.

Relevant information may include:

- symptoms and changes over time;
- laboratory trends;
- imaging and pathology reports;
- previous tests and procedures;
- treatments and whether they worked;
- medications;
- clinician notes;
- completed and pending referrals;
- family history and established risk factors;
- outside medical records;
- patient-reported information.

The system would first identify the medically reasonable next actions for a specific patient. These could include a laboratory test, imaging study, procedure, specialist referral, defined monitoring period, or collection of additional information.

It would then estimate which option is most likely to clarify the remaining diagnostic possibilities.

The exact model architecture is not fixed. It should be selected only after the clinical use case, target population, available data, outcome definition, evaluation plan, safety requirements, and hospital workflow are established with domain experts.

Model performance will depend more on representative longitudinal data, accurate outcome labels, and rigorous evaluation than on choosing the newest model architecture.

We will separate patients across training, validation, and test datasets; compare the model with clinical guidelines and conventional baselines; measure calibration and subgroup performance; document failure modes; and require the system to abstain when the available evidence is insufficient.

Research Roadmap (How We Build It)

This is a staged research project. Each phase must meet its evidence and safety requirements before the next phase begins.

1. Define & Map	Work with clinicians and researchers to select one cancer-related diagnostic pathway, define the intended users and patient population, map the current sequence of tests and referrals, identify common sources of delay, and determine what counts as a definitive diagnosis.
2. Build & Validate	Create a retrospective research prototype that reconstructs patient pathways, evaluates possible next actions, establishes clinical and non-AI baselines, and tests whether the system can recommend useful next steps on held-out historical cases.
3. Study & Integrate	With institutional approvals and clinical partners, test the system silently within a hospital environment, evaluate how its recommendations compare with real clinical decisions, measure safety and alert burden, and study whether it could fit into the existing EHR workflow.

Current status: early research and problem definition. We are seeking clinical guidance, data-access advice, and feedback on study design. We are not selling a medical product or claiming that the system can diagnose cancer or improve patient outcomes.

How the Intended Workflow Would Work

A future clinical workflow could operate as follows:

1. The system reviews the patient's available longitudinal record.
2. It identifies the main unresolved diagnostic possibilities based on the documented evidence.
3. It recognizes which tests, treatments, procedures, and referrals have already been completed.
4. It generates a set of medically reasonable next actions.
5. It estimates which action would reduce the most diagnostic uncertainty while considering risk, invasiveness, time, cost, availability, and possible duplication.

6. It presents the clinician with one recommended next step, reasonable alternatives, supporting evidence, and an uncertainty statement.
7. The clinician decides whether to accept, reject, or modify the recommendation.
8. When a new result, visit, procedure, referral, or symptom is recorded, the system automatically recalculates the pathway.

For example, the system might initially recommend repeat blood testing. If the result shows worsening anemia, it could update the pathway and prioritize gastrointestinal evaluation. If that evaluation is negative, it could redirect attention toward another appropriate specialty rather than repeating the same work.

Every update should clearly show:

- what new information was added;
- what the previous recommendation was;
- what the new recommendation is;
- why the ranking changed;
- which evidence supports the change;
- how confident the system is.

The platform should favor safety over certainty. If the evidence is incomplete, contradictory, unfamiliar, or outside the model's validated population, it should abstain rather than force a recommendation.

Why We Start With One Cancer Pathway

Cancer Compass is intended to support multiple cancer-related diagnostic pathways over time, but the first study should focus on one clearly defined clinical situation.

Possible starting areas include:

- unexplained anemia and possible colorectal or hematologic cancer;
- persistent gastrointestinal symptoms and colorectal or pancreatic cancer;
- recurrent pulmonary symptoms and lung cancer;
- persistent oral lesions and head-and-neck cancer;
- recurrent hematuria and bladder cancer.

The first pathway should be selected with hospital partners based on:

- frequency of diagnostic delay;
- clinical importance;
- availability of longitudinal data;
- number of departments involved;
- variation in current diagnostic pathways;
- whether an earlier or better-ordered action could realistically help;
- whether the final diagnosis can be reliably identified.

Beginning with one focused pathway allows us to study whether the model can reconstruct prior decisions, rank reasonable next actions, avoid unnecessary duplication, and improve diagnostic efficiency before considering broader cancer applications.

The first milestone is not a hospital-wide deployment. It is a carefully designed retrospective study with credible evaluation criteria, clinical supervision, and clear safety limits.

Privacy, Security, and EHR Access

Cancer Compass would require access to sensitive medical information. Privacy and security must therefore shape the system from the beginning.

The intended architecture would operate within a:

- hospital-controlled private cloud;
- secure hospital tenant;
- approved institutional computing environment;
- or on-premises system.

Identifiable patient information should remain inside the hospital's approved environment. It should not be sent to public AI services.

The system would require:

- role-based access controls;
- encryption in transit and at rest;
- complete audit logs;
- hospital-controlled authentication;
- minimum-necessary data access;
- written retention and deletion policies;
- controls separating clinical use from research use;
- appropriate agreements with technology vendors;
- institutional approval for any research involving patient information.

Where supported, the system could connect to the EHR through controlled interfaces such as FHIR APIs, FHIR Subscriptions, CDS Hooks, or hospital-specific event systems.

These connections could allow the patient's diagnostic pathway to update automatically when a new laboratory result, imaging report, procedure, appointment, note, or referral is recorded.

The hospital would control what information the system may access and whether any data may be used for model improvement.

Updating Clinical Knowledge

The system should remain aligned with current clinical guidance, but it should not change its medical behavior automatically whenever an external website is updated.

Potential knowledge sources may include:

- National Cancer Institute guidance;
- ASCO clinical-practice guidelines;
- ACR Appropriateness Criteria;
- licensed NCCN materials;
- hospital-approved local protocols;
- other reviewed professional standards.

Clinical updates should follow a controlled process:

1. Identify a new guideline or protocol version.
2. Determine what recommendations changed.
3. Confirm that the content can legally be used.
4. Have qualified clinicians review the proposed update.
5. Test the change on historical cases.
6. Approve the new version before deployment.
7. Preserve the prior version and complete audit history.

The system should identify which guideline, protocol, or patient evidence contributed to a recommendation whenever possible.

What This Could Mean for Patients and Clinicians

If research eventually shows that the approach is accurate, safe, and useful, Cancer Compass could help clinicians navigate difficult diagnostic cases more efficiently.

It could help determine which next step is most likely to move the patient toward a definitive answer rather than allowing the case to move through repeated or disconnected appointments.

The intended value is not simply fewer referrals. It is:

Fewer unnecessary or duplicative referrals, while helping necessary referrals happen earlier and reach the correct specialty.

The system could also reduce the amount of time clinicians spend reconstructing years of records and checking which tests or procedures have already been completed.

For patients, this could mean fewer unnecessary appointments, less repeated testing, less travel, lower cost, and less time spent without a clear diagnosis.

Potential Practical Value

- A structured view of the patient's complete diagnostic journey
- Automatic recognition of newly completed tests, procedures, and referrals
- Identification of repeated, circular, or potentially duplicative diagnostic work
- Ranking of medically reasonable next actions
- A recommendation focused on reducing diagnostic uncertainty
- Consideration of risk, invasiveness, cost, waiting time, and availability
- Earlier direction toward the most relevant specialist or procedure
- Automatic recalculation after every important clinical event
- Transparent explanations linked to the patient's record
- Clear uncertainty statements and model abstention
- A supporting patient app for symptom tracking and diagnostic progress
- Evidence that clinicians and researchers can inspect, challenge, and improve

Patient Application

The main product would be designed for hospitals and clinicians. A patient-facing application could support the diagnostic process without independently diagnosing cancer.

Patients could use it to:

- track symptoms and changes over time;
- report whether treatments helped;
- record changes between appointments;
- upload permitted outside records;
- view completed and upcoming diagnostic steps;
- receive reminders approved by the clinical team;
- understand what part of the diagnostic process is currently pending.

Patient-reported information could be added to the clinician-facing timeline when appropriate.

The app should not independently tell a patient that they probably have cancer or recommend a medical procedure without clinician involvement.

Human Oversight Remains Essential

Doctors already evaluate differential diagnoses, choose tests, make referrals, and balance risks. Cancer Compass is not intended to replace this work.

Its potential advantage is computational. It could:

- review years of information quickly;
- connect evidence across departments;
- remember every completed diagnostic step;
- compare several possible next actions at once;
- detect duplicated work;
- learn from previous diagnostic pathways;
- incorporate hospital-specific waiting times and availability;
- update continuously after new information arrives.

Clinicians would remain responsible for:

- determining whether a recommendation is medically appropriate;
- considering diagnoses the system may have missed;
- evaluating contraindications;
- understanding the patient's preferences and circumstances;
- deciding whether to order a test or make a referral;
- making the final diagnosis.

Researchers and clinicians must define acceptable performance, review errors, monitor bias and drift, and decide whether the available evidence supports any real-world use.

High-risk, uncertain, unfamiliar, or contradictory cases should always be escalated to qualified humans.

What It Is Not

Cancer Compass is not:

- a cancer diagnosis;
- a cancer probability score;
- a generic medical chatbot;
- an autonomous test-ordering system;
- a replacement for a physician;
- a system that assumes cancer is always the correct explanation;
- a tool that sends every concerning patient to oncology;
- a basic reminder or scheduling platform;
- a static clinical flowchart;
- a product designed simply to reduce referral numbers.

It is not ready for patient care, and it should not be described as clinically effective until appropriate studies support that claim.

Early prototypes are research tools and must be presented with those limits clearly.

Research Roadmap at a Glance

Clinical definition	Specific diagnostic pathway, intended users, target population, available actions, outcome definition, workflow, possible harms, and success criteria agreed upon with clinical advisors
Dataset and baseline	Representative longitudinal data, reliable final diagnoses, complete pathway reconstruction, data-quality checks, and clinical or conventional baselines
Independent validation	Held-out and preferably external evaluation, subgroup analysis, calibration, safety review, model abstention, and comparison with actual historical decisions
Silent clinical study	Institutional approvals, secure EHR integration, clinician review, predefined stopping rules, alert-burden measurement, and no autonomous clinical action
Prospective study and expansion	Credible evidence of safety and usefulness before broader deployment, additional cancer pathways, or patient-facing clinical use

Current Research Phase

We are in the early problem-definition and partnership stage.

Near-term work includes:

- interviewing clinicians from relevant specialties;
- identifying where diagnostic sequencing creates avoidable delays;
- selecting one initial cancer-related pathway;
- mapping the current sequence of tests and referrals;
- determining what retrospective data may be available;
- defining what counts as a definitive diagnosis;

- identifying medically reasonable next actions;
- establishing clinical and non-AI baselines;
- designing an evaluation protocol;
- investigating overlapping products, publications, and patents;
- identifying the privacy, ethical, legal, and regulatory requirements that must shape the project from the beginning.

The central research question is:

Can a longitudinal AI system identify a more informative next diagnostic action than standard rules or single-visit models, while reducing unnecessary steps and maintaining patient safety?

Who We Are

jiawei.j.liu@gmail.com and rowankchiang@gmail.com are students in Stanford AI4ALL's Medical AI cohort.

We are developing Cancer Compass as a long-term research effort and are seeking guidance from clinicians, medical-AI researchers, health-data experts, biostatisticians, diagnostic-safety specialists, and regulatory professionals.

Our current mentor is Akshay Paruchuri of Stanford's STAI Lab.

Next Step

We would value a short conversation with clinicians or researchers who can help us pressure-test the clinical problem, identify cancer pathways where inefficient diagnostic sequencing causes meaningful delays, and determine what evidence would be required for a responsible study.

We are looking for candid feedback, clinical guidance, data-access advice, and possible long-term mentorship or collaboration—not an endorsement of an unvalidated product.

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